THE NEW LOOP

A FUTURE HISTORY OF THE AI TRANSITION

ABSTRACT

Discussion of the economic transition driven by artificial intelligence and advanced automation has largely proceeded through uncoupled debates, with proposals such as Universal Basic Income discussed in isolation and with limited structural context. Yet the transition is likely to involve many simultaneous, interacting pressures: a declining share of income flowing through labor, changing patterns of household demand, rising concentration of capital income, fiscal strain, and risks of debt-deflation and political instability. The interactions of forces like these will determine our future, but those interactions cannot be fully understood from single-policy arguments.

Instead of presenting a final solution, this paper offers a future-history narrative designed to explore how multiple policy mechanisms may interact over time within a single economic system. Its purpose is to make the structure of the transition more explicit and understandable, showing how a coordinated policy package may address several dimensions of the problem simultaneously and providing a basis for more disciplined public discussion.

The framework brings together policies including a Social Wealth Trust, Universal Basic Income, value-added taxation with rebate, distribution-sensitive income taxation, automation-responsive corporate taxation, and mortgage-adaptation mechanisms. A companion paper presents the quantitative simulation model directly, together with illustrative output graphs, while a public Streamlit implementation will allow readers to inspect assumptions, adjust parameters, and test outcomes for themselves.

The goal is to encourage better debate by making the interacting structure of the transition visible, testable, and open to revision.

Roger Ison
roger@miximum.info

INTRODUCTION

The sudden, rapid diffusion of advanced artificial intelligence through the global economy will create crises and opportunities unlike any prior event in human history. Huge efforts are presently expended on concerns such as "alignment" and "safety," which certainly are things that matter. But the most urgent questions, which will shape real lives at scale, are economic: what happens to incomes, purchasing power, investment, and social stability when automation spreads through

most forms of work? Despite its urgency, this subject is only beginning to receive the attention it deserves.

Historically, the revenues earned from economic activity are split among labor, capital, and government. In plain terms: consumers buy goods and services; businesses use that revenue to pay workers, pay taxes, and generate profits; workers then spend and save; owners invest and consume out of profits; and the cycle continues. This is capitalism: a circular feedback system that stays in balance because the ability to produce is meaningless without the ability to buy. Wages are not the only source of purchasing power - people can also consume out of dividends, transfers, credit, and asset income - but the system must reliably generate broad, spendable income at the scale required to clear the economy's output. We call this circulation the "Old Loop".

The consequences can be severe and persistent when that loop fails. Collapsing demand propagates into collapsing production: falling sales lead to layoffs and cutbacks; layoffs reduce incomes; reduced incomes further reduce sales; investment stops; and productive capacity can be impaired for years. The Great Depression was a feedback failure so profound that even extraordinary policy efforts struggled to restore normal economic activity. My grandfather, a railroad fireman who shoveled coal into steam engines, was out of work for a decade. He traded a solid gold railroad watch for a milk cow to plow the garden and feed the family. Whatever one emphasizes as proximate causes, the durable lesson is that restart can be difficult once the link between output and purchasing power is damaged, and the consequences can spill into conflict and politics for generations.

Economically, advanced artificial intelligence is an unusually powerful form of automation. It will automate information and intellectual work first. As that proceeds, it will somewhat later also accelerate automation in physical systems, mining, manufacturing, logistics, agriculture, construction, and beyond. Automation raises productivity and lowers operating costs. But it also changes the income architecture of the economy: as software and machines substitute for labor, a smaller fraction of value added shows up as wages and a larger fraction shows up as profits and returns to capital. If ownership of that capital is concentrated, then purchasing power becomes concentrated too, even while productive capacity expands.

Corporations are the engines of modern wealth creation. Only large-scale enterprises can reliably perform research and development, design, logistics, manufacturing, marketing, distribution, and global support for complex products and services. So the issue is not to stop progress, nor to replace productive enterprise with circular services in which we merely transact among ourselves. The issue is how the feedback loop that connects production to purchasing power can remain stable as labor income shrinks, so that wealth creation can continue.

That's the core conundrum. If wages stop being the dominant channel by which most households receive income, where does broad purchasing power come from? If we want a prosperous society that continues to invent, build, and improve while much of the operational work is automated, then we will have to redesign parts of the income-and-ownership system, not just "patch" it.

This is not fundamentally a left-right political argument. It's a systems-engineering problem in the design of a modern economy. The timing of change is uncertain; the direction is not. And the failure modes are reasonably clear.

One failure mode is a concentration trap: the rents from automation accrue to narrow ownership, purchasing power becomes too concentrated, demand weakens relative to capacity, and the system's social legitimacy erodes. Another is a treadmill trap: we attempt to offset structurally declining labor income with ad hoc transfers and stimulus without changing the underlying architecture, producing fiscal strain and political fragility without restoring a durable, self-supporting loop.

There are plausible ways forward. They likely involve some combination of broadened ownership, redesigned tax bases suited to an economy where labor income is no longer central, and income mechanisms that are less tightly coupled to employment. But proposals should be judged by concrete criteria: do they keep aggregate demand sufficient to clear output, preserve incentives for investment and innovation, avoid unstable debt expansion, and remain politically sustainable over decades rather than election cycles?

This is where computational modeling becomes essential. Verbal reasoning alone quickly reaches its limits when a system is governed by feedback loops, delays, and nonlinear effects. Then ideology tends to fill the empty spaces. Models are always wrong in some respect, but a good model can be inspected, criticized, and improved. It can show where income comes from and where it goes. It can show how investment is financed, what happens as wages decline, and which policies move the system toward stability rather than crisis. It can reveal stresses and failure modes. Just as importantly, it can identify the policy regions where the system may become self-supporting again.

Current discussion of the economic impact of artificial intelligence remains largely qualitative and often proceeds one policy at a time. Universal Basic Income is a leading example. But the transition will not unfold one policy at a time; ownership patterns, fiscal flows, household balance sheets, demand, political conflict, and institutional adaptation will interact throughout.

An intuitive account of the transition and quantitative modeling are both needed.

This paper provides that intuitive account. It describes the transition as a connected process. It aims to clarify the underlying economic issues, show how a coordinated policy package may address multiple stresses at once, and bring into view the political debates and institutional frictions that may obstruct adaptation even when workable policy paths exist.

The companion paper presents the quantitative simulation model. It describes a stock-flow-consistent representation of 20,000 households whose wealth, income, revolving debt, and mortgages roughly correspond to distributions observed among families in the United States. It also includes firms, banks, and government as aggregate actors. In that model economy, automation gradually increases the substitution of capital and software for labor. Over time, value added shifts from wages toward profits.

The underlying question is simple to state, but difficult to answer: what forms of ownership and policy architecture can keep aggregate demand, investment incentives, and social stability within a viable range as automation advances?

THE MODEL TELLS A STORY

This transition will not arrive as a single shock. It will spread unevenly across sectors, punctuated by surprises, and accelerated by feedback. So the model tells a story.

Early on, call it the "first wave," not a year on a calendar, machine intelligence stops being scarce. It becomes a commodity input: embedded in software, wrapped into workflows, and eventually integrated into physical systems. Then the decisive issue is no longer whether automation is possible, but how quickly it diffuses, and what it does to the economy's income architecture.

The first visible break is in information work, not because the work disappears, but because the same output can be produced with far fewer people. Accounting and compliance become "continuous" rather than seasonal. Routine analysis compresses into automated pipelines. You still see humans in the loop, but the loop is narrower than expected and narrowing further. Productivity rises, headcount stops rising, and then quietly falls.

A partner at a large professional firm says it plainly at an industry dinner: *"We used to sell hours. Now we sell outcomes. And outcomes don't need headcount."*

Automation of the physical economy follows, slowly at first, then faster, though never as fast as in the information sector. Early robotics are specialized: systems designed for narrow tasks. But the trajectory is toward generality, toward machines that can be retrained and redeployed, because the bottleneck is no longer the actuator but the cognition that coordinates perception, planning, and control. Automation seeps into warehousing, manufacturing, mining, agriculture, construction, maintenance, logistics, and then into the parts of those domains that used to be protected by "situational complexity."

Governments and firms alike recognize that automation offers a way to overcome the labor-cost advantages that had earlier shifted so much production into developing economies and hollowed out much of the industrial base of the developed world. But this does not mean a return to the old employment bargain. China is automating aggressively and building the infrastructure to support it. This drives down the cost of industrial capability, so others feel compelled to follow. *"We had better do it too!"* becomes both a boardroom conclusion and a national policy instinct.

The result is that reindustrialization, where it occurs, needs far less labor than the industrial recoveries of the past. Output may return; broad hiring does not. Pressure on wages and employment therefore intensifies rather than eases.

By the "second wave," firms begin to treat hiring as the expensive, irreversible decision. Managers defer hiring even when business is strong, because the option value of automation is rising.

Displaced workers often find work, but less often find work that pays like the jobs that disappeared. Wage growth softens, then the wage share begins to slide.

At the same time, capital demand rises. Automation is not free. Even when cognition becomes cheap, building the intelligent equipment and software infrastructure needs up-front investments in hardware, sensors, integration, robotics, data pipelines, security, and the organizational work of changing how production is done. And because the advantage is competitive, speed matters. Firms borrow to transform. Banks and markets accommodate, until they start to notice that the same technologies raising productivity are also reshaping purchasing power.

Now the story becomes macroeconomic.

A strange squeeze develops. Households attempt to maintain consumption while labor income weakens. Some draw down savings. Some borrow. Revolving credit expands in the places you would expect: the middle and lower middle, where a loss of hours or wages is not a "temporary inconvenience" but a balance-sheet event. Firms borrow to finance automation investment. The economy's ability to produce improves, but the economy's ability to pay for what it produces starts to weaken and will collapse unless something replaces wage-linked purchasing power at scale.

Prices respond to opposing forces. Productivity and competition push prices down, especially in tradable sectors. But markup capture can push the other way, especially in concentrated markets or where switching costs are high. The model does not assume a single outcome. It lets pass-through and capture contend, because the policy regime must survive either a deflationary transition, an inflationary one, or a mixed reality.

Either way, debt service becomes the fragility channel. If the regime tilts deflationary, nominal incomes soften while nominal debts remain fixed. If it tilts inflationary, households face higher consumer prices while wages lag. In both cases, the upper tail cracks first. The warning sign is not the average household; it is the ninety-fifth percentile of debt-service pain. Lenders begin to see a pattern: this is not a normal cycle, and not a one-time shock. It is structural. It is the automation transition.

A bank risk officer puts it in a sentence that makes the room go quiet: *"The collateral is fine. The borrowers aren't. And it's not because they're irresponsible—it's because the income architecture is changing."*

Then comes the recognition moment. Credit tightens where it most hurts demand, and the economy becomes demand-constrained even while productive capacity is growing. The system can still produce, but fewer people can buy.

This is where the model begins: not at the start of automation, but at the point where the feedback loop between output and purchasing power starts to break.

The design problem is now explicit. If wages are no longer the dominant channel by which households receive income, what can replace the cash flow that supports mass consumption and social stability, without destroying incentives to invest and innovate?

The model's answer is not a single lever. It is an architecture: a set of rule-based stabilizers that shift purchasing power and ownership in ways that can scale with automation.

The first move is to broaden ownership of productive assets gradually, credibly, and with a hard cap. The model introduces a Social Wealth Trust: a trust fund held on behalf of citizens, not an operating arm of government. Its purpose is not to run companies; it's to accumulate equity ownership and distribute part of the resulting returns. The trust activates only when measurable stress crosses a threshold, so it behaves like an automatic stabilizer rather than a permanently expanding institution.

When the trigger condition is met, the trust can be kick-started with bank-created credit. The bank lends to the trust; the trust purchases a slice of outstanding equity; households receive the cash proceeds; and the trust repays the loan over time. That launch mechanism matters because it accelerates broad ownership without requiring an immediate tax grab or a politically fragile appropriation.

But the core of the ownership transition is not the launch. It is the ongoing mechanism: what the model treats as a form of equity-dilution tax that is implemented as new share issuance allocated to the trust until an explicit cap is reached.

The obvious question is: why would corporations accept that?

They accept it in this story for the same reason they accept many things that look costly in isolation: because the alternative is worse, and because the rule can be made predictable, limited, and compatible with capitalist incentives.

A CEO of an automation-heavy manufacturer says it bluntly in an earnings call, after an analyst presses on "why tolerate dilution at all?": *"We can build capacity faster than the economy can absorb it. Customers don't come from spreadsheets. If broad purchasing power collapses, we're not 'more efficient'. We're stuck with stranded capability. A stable rule we can price is better than a political firestorm we can't."*

That is the hard constraint: in a mature automation transition, the binding constraint is not productive capacity. It is demand. Firms cannot monetize productivity if the consumer base is insolvent or purchasing power is too concentrated. The dilution rule is framed as a demand-preservation instrument: it converts a fraction of the automation dividend into distributed purchasing power without requiring firms to raise wages for jobs that no longer exist.

A CFO phrases the acceptance logic in the language markets understand: *"If it's universal, capped, and rule-based, it's not 'expropriation', it's a cost of doing business in a high-automation economy, like a tax, like regulatory compliance, like stock-based compensation. The market can price it. What markets can't price is unpredictable backlash."*

That predictability is central. A small, stable, widely applied dilution rate is something investors can incorporate into valuation. It reduces the risk premium associated with tail events: windfall confiscation, arbitrary sector-specific punishments, or emergency interventions that arrive

precisely when credit is already tight. A rule-based mechanism is not "free," but managers can read and understand it, and legibility lowers the probability of chaotic, discretionary policy.

A portfolio manager adds the quieter point capital markets rarely say out loud: *"The real threat to returns isn't a few basis points of annual dilution. It's the social legitimacy gap. If the system stops working for most people, the political system will eventually reach for something much cruder."*

The mechanism is also framed as a substitute for more distortionary tools. As labor income becomes less central, the tax base must shift. Governments will come after profits, consumption, wealth, or all three. In that political reality, a capped dilution mechanism can be presented as growth-compatible: it is gradual; it doesn't depend on brittle enforcement against profit-shifting; and it builds broad claims on capital without requiring repeated political battles over explicit confiscation.

The cap matters here. The trust is not allowed to become an unlimited accumulator of control. The policy goal is to create a stabilizing ownership share, not to nationalize the corporate sector. The cap makes the mechanism credible to founders, boards, and long-horizon investors: it ensures the rule does not converge toward takeover. And the trust's role is structured as passive ownership in economic intent: broad claims on returns, not politicized management. Firms accept dilution more readily if they can believe the trust is not a steering committee.

Under those conditions, the equity-dilution rule reads less like punishment and more like a premium paid for macroeconomic stability and continued license to operate in a society where labor's role in income formation is shrinking. It's a structured way for firms to buy what they need: solvent consumers, political legitimacy, and a predictable policy environment in which long-term investment remains rational.

The trust fund accomplishes another important goal: it addresses the duration-matching requirements of citizens' income funding needs over normal human lifecycles. In today's world, people are born without assets, although they may inherit some wealth eventually. When families are young, they need money raise children, fund college educations, buy a house. These large financial demands get in the way of saving. Students in college need income support. And in old age we have Social Security, but its implementation in some ways a pyramid scheme built on the idea that the young will finance the old. None of this works in an aging, deflating economy. The trust fund is a national endowment, a property trust owned by the citizens themselves, with a perpetual horizon.

The second pillar in the story is a broad consumption tax, modeled as a VAT wedge between producer and consumer prices, paired with a targeted low-income credit ("prebate") so the tax does not function as a regressive demand-killer. A VAT is attractive in this world because it scales with consumption in an economy where wages are no longer the primary taxable flow, and because it can be enforced more reliably than an ever-more-complex wage-tax system. The model does however include an income tax (flat rate on wages above a minimum income), and an indexed corporate tax that rises somewhat as wages fall.

The third pillar is Universal Basic Income, defined in a way that matches the systems-engineering problem. In the model, UBI is computed to fill the gap between market incomes and a baseline real-income pool. It is indexed to the price level, so it stabilizes purchasing power rather than merely moving nominal transfers around. And it has a ratchet: once it rises, it does not fall. That is a realistic political assumption as much as an economic rule: benefits that can be withdrawn easily are not a stable foundation for an economy undergoing structural labor displacement.

Funding is handled as a hierarchy rather than a slogan: some share can be financed by monetary issuance; then trust balance contributes; then government balance contributes; with issuance as the final backstop. The model is not asserting that issuance is consequence-free. It is making the backstop explicit so the model can test how much is required, when it is required, and what happens if it is used too aggressively.

These mechanisms work together to support one existential objective: keep the economy operating in a viable region as automation advances. They are parameterized rules whose job is to preserve the feedback loop that makes a modern economy function: production, incomes, consumption, investment, and social legitimacy.

• • • •

If this story is right, the payoff is enormous. The automation transition fades from emergency into baseline. Productivity gains appear as rising real living standards rather than as a collapse in employment-linked income. The economy continues to invest, invent, and build because consumers remain solvent and demand remains sufficient to clear output.

Even if the story is wrong, a quantitative model of these policies is still useful because it makes failure modes visible. You can see where the loop breaks, which instrument fails first, and what kinds of adjustment might restore stability.

But whether the story so far is right or wrong is not the only question. So far it is a narrow tale, even a parochial one. How might different countries arrive a solution of this kind? What would it take? Artificial intelligence and advanced automation are global phenomena. Countries and regions will find their own paths forward, so there is also a global story.

SAME STORM, DIFFERENT SHIPS

The hardest part of adapting to intelligent automation was that while the old political and economic rules and reflexes no longer worked, each Great Power or region was compelled to make its own mistakes. Old habits were hard to break.

The storm hit everyone, but the ships' hulls were built differently. China had the engines: manufacturing scale, engineers, and command capacity. It also had a balance sheet that deflation could crack like ice. The United States had the laboratories and the venture capital, but not the steering wheel. Its politics could argue for years about who deserved the gains even as the gains arrived on schedule. Europe had the deepest social contract and an instinct to keep people whole, but it paid for that with speed and coordination.

As for the rest of the world, people who did not invent these machines could buy them. In theory, that meant they could compress history: import the tools, skip intermediate stages, and become consumer societies faster than the old powers ever could. In practice, it depended on who owned the machines, who collected the rents, and whether new productivity was converted into household purchasing power rather than exported upward and outward. The same technology that could build a middle class in ten years could also hollow out the middle class in five.

CHINA

China entered the AI era with the world's greatest manufacturing base, a deep bench of engineers, and a state apparatus that can concentrate astonishing resources. But as the first hump arrived, the early hit from automation to information-sector wages and early-career employment, Chinese youth unemployment was already in the high teens. China also carried an enormous debt stack, especially tied to property, much of it fixed nominal. In a deflationary environment, that is balance-sheet poison.

Land-transfer income and property-linked revenues had been a core funding channel for local government, most acutely at the prefecture, county, and city level, and that channel had been shrinking for years. In a deflationary regime, the local fiscal model and the debt model fail together.

Meanwhile, China's domestic consumer economy remained comparatively underdeveloped, so exports still played an outsized stabilizing role. Yet across the developed world, re-industrialization increasingly meant automation and capital deepening, not a wage boom, and therefore not an expanding consumer base for imported goods.

Beijing tried to redirect exports toward faster-growing regions, but ran into strategic pushback and geopolitical friction, especially in the Americas. And with weaker domestic consumption, "tax the consumer to keep the macro pump primed" had less leverage than it did in economies where consumption is the dominant engine.

Then there was Taiwan: smaller, wealthier and institutionally more flexible, and the owner of the most strategically indispensable export category of all, semiconductors, the engines of intelligent machines. Beijing couldn't ignore the risk that a widening performance gap between Taiwan and the Mainland might become a legitimacy problem at home. But forceful unification risked a severe U.S. reaction. Waiting for American semiconductor self-sufficiency might reduce that risk, yet the diffusion speed of AI into the knowledge economy compressed the timeline.

Also, taking Taiwan Semiconductor Manufacturing Corporation and Taiwan's entire semiconductor industry offline might offer another possible advantage: it could delay global AI industry progress while giving China a chance to catch up with ASML and TSMC in semiconductor manufacturing technology. Yet that devilish gamble would leave a clear field for Korean and American companies, and it might be a long, long time before the world would forgive and forget. The costs to China itself could easily add up to trillions of dollars.

There was no clean solution, only choices trading one kind of instability for another. Ultimately, China's limited internal flexibility got in the way of transforming itself quickly. It could push forward most effectively by continuing to do what it already knew how to do. Xi Jinping had intended his timeline to be shaped by age, career, and succession, by the need to hand a unified China to the next leader. But events moved too fast. Internally, the system was simply not yet ready for its citizens to be wealthy without being occupied by hard work, and the risks of military force against Taiwan were too great.

Of course, the Chinese were endlessly clever about *applying* AI in their economy and their systems of political control. But with so much still depending on exports, China softened its image and adjusted its outward posture, not out of conversion but necessity. It refocused its export machine on products that helped regional neighbors in Asia and partners across Africa modernize faster. Those customers needed electric power equipment, networks, transport, and factory automation. These targets would give China a second chance in the coming global cycle of physical-sector automation, but it felt like a bitter second prize.

UNITED STATES

America sailed into the storm from the opposite direction. It had the laboratories, venture capital, software talent, and cultural permission to move fast and break things. It could turn ideas into companies faster than anyone else. That was its superpower and its trap. The gains arrived first as private property, not as public design.

When automation began to compress wages beyond a few industries, the old postures and arguments awoke on cue. Conservatives insisted that displaced workers would simply find new jobs, as if history were a law of physics. Progressives wanted the winners to pay, as if punishment could substitute for a working economic engine. Neither side would admit what the numbers showed: in a mature, automated economy, labor could no longer be the primary conduit carrying purchasing power to households at scale.

Washington regarded AI as an essential domain of power and control, and tried all the familiar old policy compromises. Borrow through the transition. Stimulate on the way down, patch on the way up. Argue about taxing wealth. Squeeze health care. But the economic leak was structural, so patchwork could not hold. When wage income stops rising, household balance sheets stop rising with it. And when households stop rising, the collateral behind the entire credit system begins to soften.

The Federal Reserve understood deflation, but it could not cut interest rates below the basic fact of missing cash flow. Monetary policy could change the price of short-term credit, but it could not guarantee the incomes that made credit safe. The stress appeared where it often does in America, first in revolving debt, then in housing, where the middle class is levered. Mortgages began failing in ways that did not look like the last crisis, and housing prices began to fall, trapping families in homes they could not afford. The payment shocks were not driven by teaser rates or exotic underwriting. They came from ordinary families whose work had been made cheaper than their debt, if not eliminated.

As the mortgage markets creaked and groaned, the problem revealed its second face. Corporate credit began to wobble too. If consumers cannot buy, revenues cannot hold. If revenues cannot hold, debt cannot roll forever. The storm that began in software reached the balance sheets of the physical economy.

Only then did the argument change. It stopped being a fight about fairness and became a fight about continuity. The question was no longer about the morality of capitalism, but how it could be mechanically stable when labor was no longer the main distribution system. The crisis forced a realization that should have been obvious in retrospect. Capitalism had to be preserved by redesigning its loop, so that household purchasing power could be sustained without relying on ever-expanding labor income.

Perhaps most remarkably, it was AI itself that made the necessary policy solutions credible and acceptable. Political arguments are always qualitative, self-serving, and incomplete. But AIs that could cross-check each other, and could reason over deep new data collection systems, made it possible to understand the problem quantitatively. Human specialists and intelligent machines described and debated the modeling using vast arrays of detailed, real numbers from the real economy.

More important, AIs made that understanding accessible to everyday people. All the work on “AI safety,” reducing hallucinations, and building medical systems smarter than most physicians paid off in an unexpected way. Machines had become more trustworthy, more transparent, and in many domains more trusted than politicians.

Machines that could explain complex matters simply and honestly provided the revelations that transformed politics. Their avatars debated human politicians fluently, in real time. The smartest politicians allied themselves with machines, not to manipulate voters, but to inform each voter at his or her own level, often personally. This didn’t end propaganda, but it changed the balance. Mark

Twain is often credited with saying, “A lie can travel halfway around the world while the truth is putting on its shoes.” Now the truth, running barefoot, began to outpace the lies.

EUROPE

European governments entered the storm with three primary weaknesses.

First, Europe was not a single ship. It was an armada of different histories, interests, and institutional designs. Coordination was slow, and each capital faced its own version of the transition dilemma. The core European Union remained very different from more nationalist and illiberal states on its periphery, whose instinct was to seek local advantage rather than continental coherence. Meanwhile, the Ukraine war shifted the balance of industrial and military relevance eastward, raising the strategic weight of Poland and, later, of Ukraine itself after accession. Germany, for all its industrial prowess, no longer stood quite so naturally at the center of the European story.

Second, the euro and its governing bureaucracy had been built above all to prevent inflation, Germany’s oldest economic fear, not to confront deflation. The European Central Bank was not well designed for the kind of emergency adaptation that the new transition required. The question was whether Europe could act together quickly enough, or whether national authorities would, in practice, be forced to improvise around the edges of the common system.

Third, Europeans tended to regard disruption as a social hazard rather than an opportunity. European political systems were built to preserve legitimacy. Demography also weighed on growth. When work and wages thinned, the first question was not “who failed?” but “what must be stabilized here?” And stability, almost by definition, is not naturally forward-looking.

On the other hand, Europeans did have tools. They had mature consumer economies and value-added tax systems already in place. They had long experience with the welfare state, wage support, and reducing working hours. They had universal health care systems, and publics accustomed to bargaining over the boundary between life and work. In a sense, Europe had been rehearsing for decades the idea that working time is negotiable, even if that had something to do with the continent’s long habit of lagging the technological frontier.

The very fact that Europeans could speak openly about reducing labor gave them a head start in the argument that mattered. They understood that the real issues were how to preserve demand, how to preserve dignity, and how to keep people in their homes if mortgages began to destabilize.

THE SAINT-GUILLAUME MEETING

Even when Gaul was a Roman province, its tribes were proudly and stubbornly independent. During the Second World War, Charles de Gaulle sustained the myth, and then the reality, of French independence from a London apartment by little more than his own abrasive force of will. After the war, he gave France a new constitution and a nuclear force de frappe, both as instruments of power and as symbols of national independence and dignity. He amputated French Algeria in the same

spirit, believing that the new France could not preserve its identity while dragging a colonial appendage behind it.

That same tradition left modern France unusually well equipped for the AI transition. It possessed both the administrative machinery and the intellectual elite needed to operate that machinery. Of all the European states, France was best prepared not merely to endure the transition, but to design and guide it.

It was the French who invented the value-added tax in the first place. Though the British coined the phrase "political economy," its manifestation as *dirigisme*, a government-guided economy, was quintessentially French. France's technical elite, trained to think in systems, supplied a steady stream of administrators able to translate policy into operational reality. France's constitutional architecture, bequeathed by de Gaulle, contained levers for decisive action when needed and legitimacy for necessary innovation. And the French people, however fiercely they argued over details, were already familiar with the idea that leisure and working time are political variables. For them, the compass heading was not completely strange. It was recognizable.

The conceptual roots of the French plan had many parents and had been discussed qualitatively and piecemeal around the world. But they were integrated, made specific, and clearly articulated in the form we know today at a four-day meeting convened by the French President at *Sciences Po* in Paris. That is the popular, and not always affectionate, name for the Paris Institute of Political Studies, one of France's most prestigious and distinctive institutions. More independent than most French universities, and supported by its own foundation, *Sciences Po* was never merely a school of political science. Its curriculum mixed history, law, economics, sociology, and international relations, and its highly international student body was required to spend a year abroad.

The conference invitees came mostly from *Sciences Po's* sister institutions in French public policy, economics, law and administration, with guest speakers and observers from around the world. Notably, no American business executives were invited.

The President of France was an energetic young man, recently elected in part because of his open and empathetic acknowledgment of the difficult economy. He opened the session by noting indicators of economic distress that were rising in France, and globally, due to rising automation and the increasing flow of profits to capital instead of labor. Then, to emphasize the gravity of the situation, he read aloud the first four articles of the Declaration of the Rights of Man and of the Citizen, the revolutionary text that still stands behind French constitutional tradition.

1. Men are born and remain free and equal in rights. Social distinctions may be founded only upon the general good.
2. The aim of all political association is the preservation of the natural and imprescriptible rights of man. These rights are liberty, property, security, and resistance to oppression.
3. The principle of all sovereignty resides essentially in the nation. No body nor individual may exercise any authority which does not proceed directly from the nation.
4. Liberty consists in the freedom to do everything which injures no one else; hence the exercise of the natural rights of each man has no limits except those which assure to the

> other members of the society the enjoyment of the same rights. These limits can only be determined by law.

"I say to you," he concluded, "that our task here is not to take from the rich and give to the poor. Nor is it to invent a wholly new society. Though we meet at *Sciences Po*, our task is not primarily political. It is economic, rooted in changing patterns of capital flows. Machines are replacing human labor, and capital flows reflect that. Therefore, whatever your personal politics might be, this is a crisis of Capitalism. We are not here to replace Capitalism but to preserve it, because Capitalism emanates from rights of property and free association, and the law.

"So, this is the question before us: How may all citizens not merely envy the products of machine intelligence and automation but afford and enjoy their benefits, and live more securely and more freely under the rights proclaimed in our Declaration? This is what we are here to answer."

LA NOUVELLE BOUCLE

The Saint-Guillaume Meeting, as it is now known, produced a document in three chapters: *Diagnosis*, *Treatment Plan*, and *Health and Wealth*. The President asked that it be written for wide readership rather than as a technical memorandum. It appeared under the title *La Nouvelle Boucle* (The New Loop), and France became the first major country to implement the program it set forth.

DIAGNOSIS

Diagnosis explained that the shrinking base of jobs and wages was not an ailment that could be treated directly. It would not be sufficient in the long run, and likely not even possible, to create enough new jobs to replace all those lost to automation. Nor could the state simply tax companies to replace the lost wage income, because consumption would weaken along with labor income. With inadequate consumption, firms could not sell enough to provide an adequate tax base.

The second part of the diagnosis was even more unsettling. Companies do not exist primarily to create jobs. They exist primarily to create wealth for those who own them. Jobs are a means and an expense, not a corporate purpose. Companies that kept unneeded workers indefinitely would become competitive failures.

The deeper problem, contradicting centuries of political habit, was not that companies and their owners were accumulating too much wealth. It was that not enough people owned sufficient equity, and therefore did not share adequately in the wealth that companies created. As machine intelligence and automation became the dominant sources of productivity and wealth, a society in which most citizens lacked any meaningful claim on capital would become mechanically unstable. An important aspect of the disease, to say it again, was that the public did not own enough equity.

The third factor was that there were, in the end, not many durable secrets about artificial intelligence. The underlying technical knowledge was available to anyone who could read and build. Restrictions on the most advanced chips could slow diffusion at the frontier but would not prevent it. In knowledge-based industries especially, most competitive advantages would be fleeting.

Without durable moats, prices in sectors touched by machine intelligence would be driven steadily downward. The result would be persistent deflation.

This recital of facts was revolutionary in the context of French social and political history, but data presented at the conference demonstrated the truth of it. The shock was somewhat tempered by similar data showing that the entire developed world was beginning to experience the same slow chill.

TREATMENT PLAN

Treatment Plan began from a simple proposition: the old economic loop which circulated output, wages, investment, and demand now suffered from a progressive condition that could not be patched up. Since wages were becoming an ever-narrower channel for both consumption and taxation, a new foundation had to be built beneath both. That foundation must sustain household purchasing power, provide an adequate tax base for the State, and still leave firms with enough profit and reinvestment capacity to remain dynamic.

In other words, though we redesign the macroeconomic channels, we cannot abolish circulation. The circulation reveals whether we did it right. The New Loop might distribute purchasing power through new and different mechanisms, but value must keep moving through the whole system. Households must still be able to buy. Firms must still be able to sell. The state must still be able to tax. Capital must still be able to reinvest. If any major channel collapses, the whole organism weakens. The economic heart must still pump, and do the same amount of circulatory work, but some blood vessels must enlarge because others are shrinking.

If this were done properly, the economic system could recover its balance and sustain itself. Done thoughtfully, the transition could take place without impoverishing some people while unjustly enriching others. With the increased productivity of intelligent automation, it should be possible to make everyone better off.

The prescription therefore fell into four main parts, each addressing a distinct failure in the Old Loop.

First, it had to replace the shrinking wage-income channel. Second, it had to redirect part of the capital stream into broad ownership and public revenue. Third, prevent debt deflation from breaking households and banks. And fourth, critically, tax policies must remain simple enough to be understood, stable enough to be trusted, and fair enough to be sustained politically.

Replace the shrinking wage income channel. Wages have traditionally done two jobs at once. They have provided households with purchasing power, and they have provided governments with a tax base. Jobs will not vanish altogether, but wage income will no longer be broad enough to carry the full weight of either function. Income therefore had to be supplemented directly.

The cleanest and fairest solution was a Universal Basic Income, set at a meaningful level, for example near the bottom third of family incomes in the pre-transition economy. This would provide a stable floor beneath consumption, with dignity and social participation.

The UBI should be funded from three sources. The first was a value-added tax on all consumption, without exception, paired with a rebate for households below a minimum income threshold. Second, dividend income from a new Public Equity Trust. And third, a small contribution from general government revenues. In the event of a shortfall, the UBI could be supplemented as necessary by limited government money issuance. In the more deflationary conditions created by intelligent automation, such issuance would be modest and not inflationary.

The value-added tax would be substantial, but its burden would be offset in two ways. First, lower-income households would receive the rebate. Second, though the UBI itself would not be taxed, all other income would be taxed at a low flat rate above a protected tax-free minimum. In this way, the system would support household purchasing power while keeping the overall tax structure broad, understandable, and economically efficient.

The old wage channel would thus be reinforced with a broader, more deliberate circulation of purchasing power.

Broaden public equity ownership and public revenue. The second set of policies was designed to address the existing concentration of capital without destroying the incentives for innovation and entrepreneurship. The aim was simple: if fewer people were going to earn their living mainly from wages, then more people had to receive part of the nation's income as owners.

A national wealth fund, owned by the people, would be established for that purpose. In effect, the nation would plant a public orchard inside the private economy. It would acquire a bounded grove, large enough to bear real fruit, but still clearly short of control. It would not seize the whole forest. This was not novel in France, which already had its FRR, the *Fonds de réserve pour les retraites,* which invested in market assets.

The fund would begin with a substantial but limited acquisition of equity. The banking system, backed by the government, would lend it a sum equal to 10 percent of the outstanding equity of existing French corporations. That loan would be used to purchase shares in each corporation in proportion to its existing equity. Dividends from those shares would be used to repay the loan. This pay-back would somewhat delay the benefits, but still give the public an immediate foothold in the productive wealth of the country.

At the same time, companies would be required to issue new dilutive shares to the national fund, equal to 2 percent of their equity at the beginning of each fiscal year. But this process would be capped. By law, the fund could never own more than 49 percent of the total equity of French companies, and the expectation was that it would not need to rise above roughly 25 to 30 percent. After the original acquisition loan was repaid, dividends flowing to the fund would become a continuing source of support for the UBI.

This would be a kind of wealth tax, but not a direct personal one. It socialized part of the capital stream without abolishing private property or private control. In this way, the public stake would widen gradually rather than through confiscation, growing along with the corporate sector itself.

The corporate tax system would also be redesigned. Undistributed profits, after depreciation and genuine capital reinvestment, would face a new tax. Distributed dividends, by contrast, would be taxed only in the hands of recipients, not at the corporate level. Dividends passing through the national fund would remain untaxed there as well.

A second tax change would tie the corporate tax rate to the overall national decline in wage income. As labor's share of national income fell, the tax rate on profits could rise correspondingly. A firm that became more productive through intelligent automation, while contributing to a shrinking wage base, could therefore afford to pay more tax on its profits. But corporations could also reduce that burden through further productive investment and through dividend distributions. The tax system, in other words, had to follow the new river of value.

Over time, that reinvestment would generate more wealth, higher productivity, and larger dividend streams, some of which would flow through the national wealth fund and the UBI to support consumption.

Protect both households and lenders from debt deflation. Long-term fixed debt, such as mortgages, requires special attention in a deflationary economy. Prices and wage income can fall even as real economic output rises. If prices fall and wages follow, a household's fixed mortgage payments become an increasing share of the monthly budget. In time, that becomes insupportable. Meanwhile the bank receives a windfall, until the mortgage fails, because the fixed nominal payment is worth more in deflated euros.

The solution was innovative and subtle. It introduced a kind of pressure regulator. By law, the original contract payment would become indexed to a hybrid measure reflecting both the price level and household income. If the economy deflated, the required mortgage payments would shrink. But the bank would still be made whole, because the government would neutralize the lender's nominal loss by paying the difference directly to the bank.

The mortgage would remain a real obligation, still a contract. But it would no longer be a rigid conduit through which falling prices and falling wages could freeze prudent households in place, force them into default, and then transmit the shock onto bank balance sheets.

This new policy was easy to defend to citizens, to the banks, and to the French state itself. It would preserve household solvency, as well as the credit system on which savings, lending, and investment depend. The duty of the state was not to side with either the creditor or the debtor, but to regulate the pressure between them so that both could pass through the transition intact.

A fair and simple tax system. The conference recommended a new tax system conceived "in concert" with the new economic loop. The authors argued that simplicity would make the long transition into this new economy easier to measure and harder to exploit or manipulate politically.

The elements of the tax proposal were these. UBI would not be taxed at all. Value-added tax would be applied in the familiar way, but rebated to low-income households, so that the rebate phased out as income rose. Income tax would be levied at a flat rate on all income above a universal minimum, with that minimum indexed to deflation. The corporate tax rate would be governed by a hybrid index of prices and declining wages. The principle was simple: as companies eliminated labor, they could afford to pay a higher rate, and the revenue would help support consumption.

HEALTH AND WEALTH

The conference understood from the outset that this transition would be long, perhaps twenty years or more. Emergency relief or a temporary correction would not be enough. France would need a social contract that could endure and stabilize a prolonged passage into a more automated economy, while remaining politically legitimate, economically intelligible, and transparently measurable in public.

This was neither socialism nor an attempt to replace the market. Companies would remain private, competitive, and strongly oriented toward profit and reinvestment. But because all citizens would collectively own the national wealth fund, the gains from corporate success would no longer flow only to a narrow class of private owners. They would also return, in part, to the citizenry as a whole.

That changed the political meaning of profit. In the old language, labor and capital stood opposed, and rising profitability could easily be seen as coming at labor's expense. The new arrangement would soften that tension, because citizens would participate not only as workers and consumers, but also as owners. Ownership changes one's perspective. Productivity, innovation, and profitability would no longer appear simply as corporate victories. They would strengthen a common national asset in which every citizen had a stake.

This new economic loop depended in part on dividend distribution, and the tax system was designed accordingly. Distributions would not be taxed at the corporate level, not as a gift to companies, but as an incentive created on behalf of shareholders, among whom the public fund would become increasingly important. UBI, meanwhile, was more than a pension substitute. It would support households across the life cycle: in youth, in study, in child-rearing, in interrupted work, and in old age. And because the system strongly favored reinvestment for corporate growth, citizens would gain not only income, but wealth. Over time, the national fund itself would grow.

The authors, while delicately avoiding certain words, presented this as a cure for the old French habit of treating profit and social protection as opposing principles. If the tax system remained simple, if dividend flow was encouraged, and if citizens came to see themselves as owners as well as workers, then growth and security would no longer stand against one another. They would become responsive elements of the same loop.

During the decades-long transition, the new loop would have to be measured openly as it evolved. Progress could not simply be proclaimed; it would have to be tracked, explained, and adjusted as household income, prices, investment, dividend flows, tax receipts, and the national fund changed. Machine intelligence could help by turning large, messy data into usable forecasts and policy stress

tests, and by presenting results in forms citizens could understand. In a transition of this length, trust would rest not only on outcomes, but on the visible, auditable integrity of the measurements.

European Union law imposed important design constraints on any French version of the New Loop. The national wealth fund could not be simply an arm of the state, exercising discretionary political control over private firms. It would have to be constituted as an independent, universal, and largely passive investment trust, governed by law and fiduciary rules rather than by ministers. Likewise, mortgage indexation and bank make-whole provisions would have to be framed as general financial-stability and social-cohesion measures, applying by rule rather than by favoritism. These constraints did not make the plan impossible, but they did require that it be designed and implemented with institutional discipline.

AND THE CYCLE TURNED

Historically, the American revolution came before the French one. The United States were first. But in this newer revolutionary age, France saw more clearly than we did what had to be done. France acted first, with a coherence, elegance, and élan that America lacked. There were rough passages, of course. As the simulations had predicted, the great deflation could not be entirely prevented, and economic policy moved, as it always does, with a lag. Yet the larger fact remained: it worked. The transition was not painless, but it held, and the world that emerged was better and fairer than what might so easily have followed.

Income inequality eased, though the wealth span remained. The bottom moved up. Households with great wealth did not lose that wealth, but ordinary households became much less fragile. Families began saving earlier and could invest more confidently in education. Universal Basic Income didn't end work, but it was a gift of freedom. People could pursue ideas, start enterprises, start families with confidence, and some could choose work for its meaning as well as its income. For those so inclined, work became less a matter of compulsion and more a matter of vocation.

And these changes made capitalism stronger. Consumers remained solvent. Demand remained real. Firms kept investing because markets still cleared. Innovation could still be highly rewarded. But the system no longer depended so heavily on fragile households, debt expansion, and chronic insecurity. Incentives were better aligned. Ownership was broader. Social stability was stronger.

It might not have turned out that way. But it did.

March 6th, 2026.

That future history is only one possible scenario. We can't know the speed or full extent of the transition in advance. Artificial intelligence and advanced automation will move faster in some sectors than in others. Bottlenecks, delays, and political resistance are also likely. Many indications suggest that events in Taiwan could be less benign than my story.

But uncertainties are exactly why we need computational modeling. A parameterized and adjustable framework allows different rates of change, different depths of impact, and different policy responses to be examined directly.

I began this project about four weeks ago. Two of those four weeks were spent writing and polishing this scenario paper and its companion. Today the New Loop model consists of 6,345 lines of Python code. I wrote none those lines myself, though I closely guided their incremental development. The project does include one utility module, 863 lines long, that I wrote 12 years ago in C#. OpenAI's Codex translated it to Python flawlessly in a few minutes.

ChatGPT prepared a deep-research review of the final version of this paper in about six minutes. It took me much longer than that to read the review and think through its implications. That review is also available as a separate paper, *ChatGPT Critique of the New Loop Scenario*. Some readers might find it useful for the references cited.

Three months ago, in December, this would not have been feasible. That is information-sector automation happening in real time.

When I started, it was unclear to me what this model might reveal. Building it was a discovery experience. Revolutions and great wars have emerged from much less disruption than what AI may bring, but the modeling leaves me optimistic. There are plausible paths forward that do not leave workers behind. Capitalism won't be abolished. It is not inevitable that wealth and power will concentrate still further while impoverishing most of society.

This is an opportunity for a politico-economic reset that we have needed for a long time.

Politically, there is no easy path. Expect endless crippling argument. Conservatives will cling to inherited structures. Progressives will be tempted to expropriate wealth. Corporations and wealthy stockholders would fiercely resist dilution, even if it is slow, measured and capped. Some countries, shaped by different histories, institutions, and civic habits, will navigate this transition better than others.

Still, my sense is that the New Loop is structurally sound and directionally correct. I spent much of my adult life investing against inflation, and there may yet be more of that. But intelligent automation is likely to be deflationary. Deflation can be slow and exhausting, or devastating. People displaced by automation will often find it very hard to find new work. For many older workers, it may be impossible. No modern economy can remain healthy by shifting everyone into marginal personal services.

The cash flows of economies everywhere will have to be rewired. America's Social Security system was built on the assumption that current workers support retirees through current wage taxation. It is already under strain. Large-scale job losses from intelligent automation would break that structure.

Nor will it be possible for the federal government simply to spend its way out of the dilemma. Wealth taxes won't solve it either. There is not enough accessible wealth for that to work on the required scale, and untethered wealth taxation would damage saving itself. It would push people who had acted prudently over a lifetime into unnecessary dependence.

I hope this work has engaged and informed you, and that some readers will be tempted to read the companion paper and learn about the New Loop simulator. I also hope it contributes to a clear-thinking debate about the most consequential technology ever created.